Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit. Questions refer to the nonindented paragraph above unless otherwise stated.

Question permutation table:

Version		1	2	3	4
Question					
1	→	3	1	3	1
2*	→	4	2	4	2
3	→	1	3	1	3
4*	→	2	4	2	4

*: Question follows previous

According to the 2019 U.S. Census Bureau estimates, 63.4% of the U.S. self-identifies as Non-Hispanic white, 15.3% as Hispanic and Latino, 13.4% as Black or African American, 5.9% as Asian, and 2.0% as Other. A random sample of 1000 students graduating from California colleges and universities resulted in the following data on self-identified race:

Self-identified Race	Non-Hispanic White	Black or African American	Hispanic or Latino	Asian	Other
Number in Sample	679	51	77	190	3

Use this information to answer questions 1 and 2

1. In the null hypothesis, what does the symbol μ_{Race} represent?

   (a) The proportion of the U.S. that self-identify as a certain Race
   (b) The proportion of Californians that self-identify as a certain Race
   (c) The proportion of all students graduating from California colleges and universities that self-identify as a certain Race
   (d) The proportion of the sample of 1000 students that self-identified as a certain Race

   Correct answers: (c)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(d)	(c)	(a)	(d)
(b)	→	(c)	(b)	(c)	(a)
(c)	→	(a)	(d)	(d)	(b)
(d)	→	(b)	(a)	(b)	(c)

2. Based upon a p-value of less than 0.001, what is an appropriate conclusion for the goodness-of-fit test?

   (a) Strong evidence does not exist that the distribution of self-identified race among California college and university students differs from the distribution of self-identified race in the U.S. overall.
   (b) Strong evidence exists that the distribution of self-identified race among California college and university students differs from the distribution of self-identified race in the U.S. overall.
   (c) Strong evidence exists that the distribution of self-identified race among California college and university students is the same as the distribution of self-identified race in the U.S. overall.
   (d) Strong evidence exists that the probabilities of each self-identified race among California college and university students are not all equal.

   Correct answers: (b)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(c)	(d)
(b)	→	(d)	(c)	(b)	(a)
(c)	→	(c)	(a)	(a)	(b)
(d)	→	(b)	(b)	(d)	(c)

Are people equally likely to be born on any day of the seven days of the week? Or are some days more likely to be a person's birthday than other days? To investigate this question, days of birth were recorded for the 147 "noted writers of the present" listed in The World Almanac and Book of Facts 2000. The counts for the seven days of the week are given in the following table.

Mon	Tues	Wed	Thurs	Fri	Sat	Sun
17	26	22	23	18	15	25

Use this information to answer questions 3 and 4

3. Are validity conditions met to conduct a theory-based chi-square goodness-of-fit test?

   (a) Yes, since all observed counts are at least 10.
   (b) Yes, since we have at least 10 successes and 10 failures.
   (c) No, since the p-values in questions 35 and 37 were different.
   (d) No, since this was not a random sample.

   Correct answers: (a)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(b)	(a)	(d)
(b)	→	(c)	(d)	(b)	(a)
(c)	→	(d)	(a)	(d)	(c)
(d)	→	(b)	(c)	(c)	(b)

4. Based upon a p-value of 0.53, what is the appropriate conclusion for the test?

   (a) Strong evidence exists that people are equally likely to be born on any day of the seven days of the week.
   (b) Strong evidence does not exist that people are equally likely to be born on any day of the seven days of the week.
   (c) Strong evidence exists that people are not equally likely to be born on any day of the seven days of the week.
   (d) Strong evidence does not exist that people are not equally likely to be born on any day of the seven days of the week.

   Correct answers: (d)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(b)	(b)	(b)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(a)	(d)	(a)	(b)
(d)	→	(c)	(a)	(c)	(c)